MIL-I-50867B (PA)
12 July 1976
SUPERSEDING
MIL-I-50867A (PA)
17 April 1975

MILITARY SPECIFICATION

IGNITION ELEMENT ELECTRICAL LOADING, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 This specification covers the loading, assembling and packing of the Ignition Element Electrical intended for use with 152MM cartridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-STD-650 - Explosive: Sampling Inspection and Testing

DRAWINGS

Product and Package Drawings

9249026 - Sub-Assembly
9249028 - Ignition Element Electrical
9249226 - Box, Setup Packing, Ammunition Inner for Initiator Assembly
9204227 - Box, Setup, Packing Ammunition Outer For Initiator Assembly
9204228 - Box, Packing, Ammunition for Initiator Assembly

FSC: 1375

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3. REQUIREMENTS

3.1 Materials, Ingredients, Drawings and Processes.-Materials, Ingredients, drawings and processes shall be in accordance with applicable drawings, specifications and quality assurance plans.

3.2 Sub-Assembly

3.2.1 Welds.-The welds of the sub-assembly bridge ribbon shall support the weight of the sub-assembly without any weld breaking.

3.2.2 Circuit Continuity.-The electrical resistance of the sub-assembly shall not be greater than 0.50 ohm or less than 0.20 ohm (collar to cup)

3.3 Ignition Mix

3.3.1 Ignition Mix Preparation.-(see 6.7). The contractor shall prepare an in-process quality plan. The plan shall assure that wet solvent mix, from assembly of ingredients to loading of ignition mix into sub-assembly 9249026, is processed and handled in such a manner that it will assure compliance with uniformity requirements of paragraph 3.4.1 and drawing 9249028.

3.3.1.1 Weight Tolerances.-Ingredients for the Ignition Mix shall be accurately weighed on a precision balance and shall conform to the requirements of Table I. Volumetric measurement of ingredients is not acceptable.

TABLE I

Potassium Chlorate	40% ± .1%
Lead Thiocyanate	32% ± .1%
Charcoal	18% ± .07%
Lacquer	10% ± .05%

Note: The purpose of Table I is to supplement drawing 9249028 and facilitate compliance with end item requirements of drawing 9249028 and uniformity paragraph 3.4.1.

3.3.1.2 Compliance.-Following contractor verification that the in-process quality plan has produced acceptable item, the plan shall be submitted to the cognizant government technical agency for review. The on-site government inspector shall verify by visual inspection, at random intervals, that contractor personnel are in compliance with the in-process quality plan.

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3.4 Ignition Element Electrical

3.4.1 Ignition Mix Uniformity (see 6.9).-The uniformity of the ignition mix in the completed Ignition Elements Electrical shall conform to drawing 9249028 when tested in accordance with Paragraph 4.5.5.

3.4.2 Circuit Continuity.-The electrical resistance of the ignition element electrical shall not be greater than 0.50 ohm or less than 0.20 ohm (collar to cup). (See 6.14)

3.4.3 Non-Functioning.-The ignition element electrical shall not function when energized with a direct current (d.c.) of 0.3 amperes for 0.3 seconds. This test shall not cause broken or torn leaves.

3.5 Workmanship

3.5.1 Parts and Assembly.-All parts shall be fabricated and finished in a workmanlike manner. They shall be free of chips, cracks, dirt, grease, flux, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All assemblies shall be fabricated such that no components are free or loose.

3.5.2 Ignition Mix.-The completed ignition mix shall be free of dirt, cracks, fissures, porous material and other defects which would make the item unsuitable for the intended use.

3.6 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made apart of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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TABLE III FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Ignition Element Electrical and Components				9249028
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	Material, Ingredients, drawings and processes			3.1	4.5.1
	Sub-Assembly				
	Strength of bridge ribbon weld	50		3.2.1	4.5.2.1
	Circuit Continuity	50		3.2.2	4.5.2.2
	Ignition Mix Preparation	-		3.3.1	4.5.3/6.6.1
	Bridge ribbon broken or multiple bridge ribbon	100		4.4.2.2	Visual/4.5.1.1
Informational	Calorimetry Test	30 gms		3.1	4.5.9/6.11
	Lacquer Solids	-		3.1	4.5.7/6.8
Major A	Ignition Element Electrical				
	Ignition Mix Uniformity	100		3.4.1	4.5.5/6.9
	Circuit Continuity	100		3.4.2	4.5.4.1
	Non-Functioning	100		3.4.3	4.5.4.2

NOTES:

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TABLE III FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Continuation Sheet	2 OF 2	9249028
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Hole in outer cup not aligned with hole in front support	100		4.4.2.1	Visual/4.5.1.1
	Torn leaves on outer cup	100		4.4.2.3	Visual/4.5.1.1
Informational	Ignition Charge Composition Mix	6		3.1	4.5.6/6.11
	Closed Bomb Tests	10		3.1	4.5.8/6.11
5 Minor	Evidence of Poor Workmanship	100		3.5.1 3.5.2	4.5.1 4.5.1.1

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Support Front Assembly		9249027
			NEXT HIGHER ASSEMBLY 9249026

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Evidence of exposed adhesive		100%	3.1	Visual/4.5.1.1
102.	Tabs of collar not folded		100%	3.1	Visual/4.5.1.1
Minor					
201.	Evidence of poor workmanship		.65%	3.5.1	Visual/4.5.1.1

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9249026
4.4.2.2	Sub-Assembly				NEXT HIGHER ASSEMBLY 9249028
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Bridge ribbon broken or multiple bridge ribbon		*.65%	3.1	Visual/4.5.1.1
102.	Hole in outer cup not aligned with hole in front support		*.65%	3.1	Visual/4.5.1.1
103.	Overall length max.		.65%	3.1	Gage
Minor					
201.	Evidence of poor workmanship		.65%	3.5.1	Visual/4.5.1.1

NOTES: *The first article and first five lots shall be inspected 100%.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER 9249028
4.4.2.3	Ignition Element Electrical				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Material, Ingredients, Components and process			3.1	4.5.1/4.5.1.1
102.	Ignition Mix Uniformity		100%	3.4.1	4.5.5/6.9
103.	Circuit Continuity		100%	3.4.2	
104.	Non-Functioning		1.0%	3.4.3	
105.	Outside Diameter (max)		1.0%	3.1	Gage
106.	Ignition charge in slotted area above flush of front support or outer cup		1.0%	3.1	
107.	Ignition charge below flush of front support		100%	3.4.1	4.5.5
108.	Evidence of adhesive on perforated end of outer cup		1.0%	3.1	Visual/4.5.1.1
109.	Torn leaves on outer cup		1.0%	3.1	Visual/4.5.1.1
Minor					
201.	Evidence of poor workmanship		.65%	3.5.1 3.5.2	Visual/4.5.1.1
Informational					
	Lacquer solids	Each lot		3.1	4.5.7/6.8

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9249026
4.4.2.4	Box, Set-Up Inner				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major A					
101.	Contents of inner box loose		.40%	3.1	Manual/4.5.1.1
Minor					
201.	Incorrect marking		.65%	3.1	Visual/4.5.1.1

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-50867B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9204228
4.4.2.5	Box, Packing For Ignition Element Electrical				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Packing box damaged with contents exposed		.65%	3.1	Visual/4.5.1.1
102.	Strapping missing broken or loose		.65%	3.1	Visual/4.5.1.1
103.	Box improperly sealed		.65%	3.1	Visual/4.5.1.1
104.	Contents loose		.65%	3.1	Visual/4.5.1.1
Minor					
201.	Incorrect marking		1.0	3.1	Visual/4.5.1.1
202.	Evidence of poor workmanship		1.0	3.5.1	Visual/4.5.1.1

NOTES:

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4.3 First Article Inspection (see 6.6)

4.3.1 Submission.-The contractor shall submit an initial production lot sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The initial lot sample shall consist of 30 grams of cured ignition charge composition mix, 50 sub-assemblies and 100 ignition elements electrical, which have been produced by the contractor, or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment, which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, and processes. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawings, specification or source of supply as to significantly affect product uniformity as determined by the government. (see 6.10)

4.3.2 Inspections to be Performed.-First Article samples will be subjected by the Government to any or all the requirements of the applicable drawings and to any or all of the examinations or tests specified in 4.4.2 and 4.4.3 of this specification.

4.3.3 Rejection.-See MIL-A-48078 (PA)

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Lot numbering is required. Lot numbering shall be in accordance with MIL-STD-1168.

a. Each lot shall contain ignition mix having constituent materials from one lot (see 6.4).

b. Batch formation.-An Ignition Element Electrical batch shall be defined as one containing Ignition Elements Electrical which have been loaded with ignition mix made from one homogeneous mixing operation. The mixing operation shall be carefully monitored during weighing and blending and controls established to prevent settling or separation of ingredients during loading operations. (see Para (3.3.1 and 3.4.1) A batch of ignition mix shall not be retained for more than 7 days.

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4.4.2 Examination.-See MIL-A-48078.

a. Classification of defects and test tables sampling plans for major and minor defects listed in the tables starting with first article through paragraph 4.4.2.5 shall be in accordance with MIL-STD-105 Inspection Level II.

4.4.3 Testing.-Testing is described in the First Article Tables and in the Quality Conformance Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with First Article through paragraph 4.4.2.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6 herein.

4.5 Test Method and Procedures

4.5.1 Materials, Ingredients, Components and Processes.-Compliance with all requirements of section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, ingredients and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of sections 3 and 5 of this specification in accordance with 4.4.2 and first article requirements contained herein.

4.5.2 Sub-Assembly

4.5.2.1 Strength of bridge ribbon weld.-This test shall be conducted using government approved test procedures and equipment in accordance with paragraph 6.3.

4.5.2.2 Circuit Continuity.-The sub-assembly shall be tested for electrical resistance by applying a direct current of 10 milliamperes maximum using a probe through the outer cup hole to the inner sleeve or collar as one contact point. The outer cup shall be used as the second contact point. Assure that the probe and all sub-assembly parts are free of dirt, fingerprints, moisture, grease, etc. before attempting this test.

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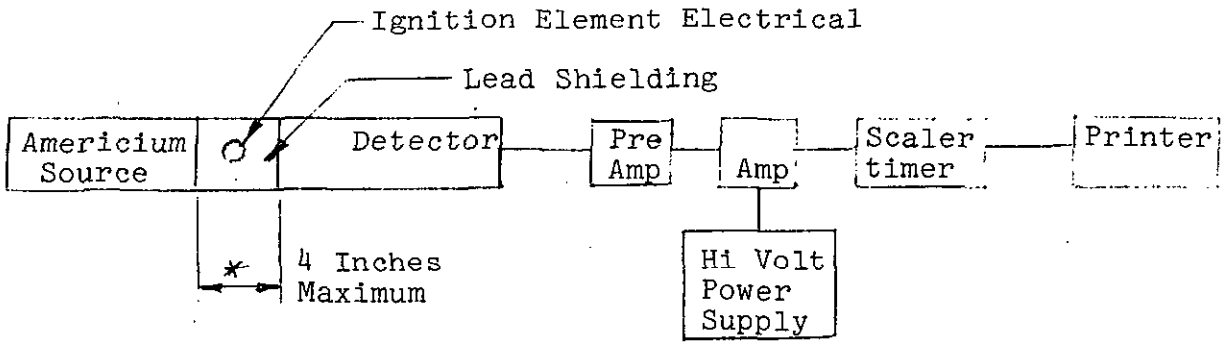
4.5.3 Ignition Mix Preparation (6.7).-The ignition mix shall be prepared by individual weighing of ingredients on an approved precision balance. The dry ingredients shall be blended in an approved mixer. Blending of dry mixes with Lacquer and Butyl Acetate shall be performed using approved laboratory mixing procedures.

4.5.4 Ignition Element Electrical

4.5.4.1 Circuit Continuity.-The Ignition Element Electrical shall be tested for electrical resistance by applying a direct current of 10 milliamperes maximum using a probe through the outer cup hole to the inner sleeve or collar as one contact point. The outer cup shall be used as the second contact point. Assure that the probe and all samples are free of dirt, fingerprints, moisture, grease, etc. before attempting this test.

4.5.4.2 Non-Functioning.-The Ignition Element Electrical shall be tested for non-functioning by applying a direct current of 0.3 amperes for 0.3 seconds across the firing probe (see 3.4.3)

(4.5.5 Ignition Mix Uniformity.-Uniformity of the ignition mix shall be determined utilizing a Gamma Ray Densitometer (see 6.9). Measurement equipment shall consist of a radiation source (Americium 241, 59.2 KEV with an activity level of 100 ± 10 millicuries or an approved substantial equivalent*). Counting time shall be 2 seconds ± .01%. The measurement window for the measurement of the density of the ignition mix shall be .250-.050 in by .050-.020 inch and positioned so as to reject elements not conforming with the .0625 ignition mix maximum permissible shrink cavity depth requirement of drawing 9249028. Radiation analysis and measurement equipment shall consist of a detector, Pre-Amplifier, Amplifier, Scaler, timer, high voltage power supply, adequate shielding equipment and an automatic reader printer to record gamma measurement values.



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*Note: The distance and the activity level of the Americium source are interdependent and are critical to the effective performance of the Gamma Measurement System. A distance of two inches is recommended (see 6.13)

4.5.6 Ignition Mix Composition.-The following test methods shall be used to determine the chemical composition of the ignition mix. The percent potassium chlorate shall be computed by difference.

4.5.6.1 Charcoal.-Approximately 1.5 grams of the sample shall be transferred to a Gooch crucible containing a heavy asbestos mat. Contents of the crucible shall be extracted with five 20 ml portions of acetone saturated with potassium chlorate and lead thiocyanate. The first wash is allowed to drip through a 5-10 minutes into a 30 ml beaker. The crucible is then placed in a holder on a suction flask and the contents of the beaker poured through the crucible using suction. After the fifth wash, the crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and then dried in a vacuum oven maintained at 55 plus or minus 2 degrees Centigrade for two hours. The crucible and contents shall be cooled in a desiccator and weighed. Using a Fisher high-form filtrator or equivalent extractor, the sample shall be extracted with ten 25 ml portions of a 5 percent potassium nitrate solution followed with 50 ml of distilled water. A glass rod shall be used to break up any lumps in the sample during the extraction process. The filtrate shall be reserved for the determination of lead thiocyanate. The crucible and contents shall be aspirated for several minutes, then placed in an oven and maintained at 100 plus or minus 5 degrees Centigrade for two hours. The crucible and contents shall be cooled in a desiccator and weighed. The percent charcoal shall be calculated as follows:

$$\text{Percent charcoal} = \frac{100A}{B}$$

Where:

A = The weight of sample after acetone and 5 percent potassium nitrate solution extraction

B = The weight of sample after acetone extraction.

4.5.6.2 Lead Thiocyanate.-Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3 N nitric acid (nitric acid used should be water white and free of oxides of nitro-

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gen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AE - CD)}{B}$$

Where:
A = Volume of silver nitrate solution used.
B = Weight of the sample after acetone extraction.
C = Volume of the potassium thiocyanate solution used.
D = Normality of potassium thiocyanate solution.
E = Normality of the silver nitrate solution.

4.5.7 Lacquer solids.-Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filtering crucible and allowed to air dry for about an hour. The crucible and contents shall be placed in a vacuum oven for two hours maintained at a temperature of 55 plus or minus 2 degrees Centigrade. The crucible and contents shall be cooled in a desiccator and weighed. The procedure shall be repeated until the change in weight does not exceed 0.5 milligrams. The percent lacquer solids shall be calculated as follows:

$$\text{Percent lacquer solids} = \frac{100\ (A - B)}{A}$$

Where:
A = Initial weight of sample
B = Weight of the dried sample

4.5.8 Closed Bomb Testing.-Place the assembled Ignition Element in a Parr combustion bomb or approved equivalent. Ignition shall be effected electrically by means of an ignition wire suspended from the electrodes of the head of the bomb and in contact with the ignition element. Ignition energy shall be supplied through a 12 volt ignition box. The Combustion Bomb shall be equipped with a pressure transducer to monitor the rate of pressure buildup when the Ignition Element is ignited. The ignition mix signal shall be amplified and recorded on a memo type oscilloscope. The pressure trace of the ignition mix shall be photographed or recorded. Time shall be calculated from the difference between the 5% to 95% levels of the pressure rise trace. Ignition delay time shall be calculated from firing pulse initiation to the 5% level of pressure rise.

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4.5.9 Calorimetry Test.-The heat of reaction of the ignition mix sample shall be determined in an adiabatic calorimeter. Place a weighed amount of sample (approximately 1.5 grams) in a porcelain crucible attached to the crucible holder in the Parr Combustion Bomb. Assemble the bomb in a weighed water bucket placed in a calibrated Parr adiabatic calorimetic system. Combust the sample electrically and measure the energy liberated with a thermometer. The final reading is obtained when the temperature of the inner water bucket is in equilibrium with the outer water jacket. The heat of reaction of the sample is calculated by the following formula:

$$\Delta H_R = T\ E/W$$

Where:

ΔH_R = Heat of Reaction, cal/gm
ΔT = Temperature difference, °c
E = Energy equivalent of calorimeter, cal/°c
W = Weight of sample, gm

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level C.-Ignition Element Electrical shall be preserved and packaged as specified in drawing 9204226 and 9204227.

5.2 Packing

5.2.1 Level C.-Ignition Element Electrical packaging shall be as specified in drawing 9204228.

5.3 Marking.-Ignition Element Electrical packaging and packing shall be marked in accordance with drawings 9204226, 9204227 and 9204228.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use in 152MM Cartridges.

6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Designs For Approval.-See 6.2.3 of MIL-A-48078 (PA) Submit equipment designs as required to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

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6.4 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Submission of Certificates of Conformance and Ammunition Data Card.-In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all certificates of Conformance and Ammunition Data Cards shall be forwarded to each of the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P and SARPA-AD-E-P, Dover, New Jersey 07801.

6.7 Submission of weight results.-The contractor shall submit data of actual weights of each ingredient for each lot of ignition mix. These results shall be certified by the responsible company official and mailed in accordance with paragraph 6.6.

6.8 Egyptian Lacquer.-A lot of Egyptian lacquer shall be considered to be one shipment of homogeneous material from the same manufacturer.

6.9 Reporting Testing Results.-The results of all informational testing and gamma testing shall be forwarded in accordance with paragraph 6.6.

6.10 Submission of first article sample.-Instructions as to the location for the evaluation of first article sample shall be obtained from the contracting officer, upon request, All inquiries should be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-I-P, Dover, New Jersey 07801.

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6.11 Testing Responsibility.-First article testing for chemical composition, closed bomb testing and calorimeter testing in accordance with 4.5.6, 4.5.8 and 4.5.9 shall be conducted by the government at government expense.

6.12 Shipping Instructions.-Samples of Ignition Elements Electrical, six (6) for chemical analysis and ten (10) for closed bomb testing, shall be selected along with ten (10) samples of three (3) grams each of ignition mix. These samples shall be properly identified, packed and shipped to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-I-Q, Dover, New Jersey 07801.

6.13 Uniformity Calibration (see 3.4.1),-The contractor shall prepare ten (10) calibration standards of the Ignition Element Electrical. Five (5) standards shall conform to the minimum density limits of Table IV and five (5) to the maximum density limits of Table IV. One set of standards shall be shipped to Picatinny Arsenal in accordance with shipping instructions of paragraph 6.12. Standards shall be prepared such that they are capable of calibration of the Gamma Ray Densitometer within the minimum density and maximum density limits permitted by Table IV and drawing 9249028. Solvent-wet, ignition mix weight percentages are as follows:

Table IV

	Minimum Density	Maximum Density
Potassium Chlorate, %	39.5	40.5
Lead Thiocyanate, %	30.0	34.0
Charcoal, %	19.5	16.5
Lacquer, %	11.0	9.0
	100%	100%

Allowable weight tolerances are ± 1 milligram.

Calibration values taken from the standards shall be such that no overlap of values exists when the maximum and minimum density standards are subjected to thirty readings each. This separation of values is critical to the effective operation of the Gamma system and is dependent on the proper selection of relationships between detector distance and activity level of the radiation source (see 4.5.5).

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6.14 Circuit Continuity.-The requirement for circuit continuity applies to the sub-assembly and the completed ignition element. Inspection at both configrations is required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A173

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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